

DSC 540 - Data Preparation

Week 11 Term Project: Milestone 5

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Merging the Data and Storing in a Database/Visualizing Data

Import Required Libraries & Database Connection Setup

In this step, the required Python libraries are imported to work with the database and perform data analysis. The `sqlite3` library is used to connect to the SQLite database file, `pandas` is used to handle data in DataFrame format, and `matplotlib.pyplot` is used to create visualizations later in the project.

A connection to the SQLite database file named **manoj_project.db** is then created. After establishing the connection, a **cursor object** is created. The cursor allows SQL queries to be executed on the database.

A **try-except block** is used to handle potential errors during the connection process. If the connection is successful, a confirmation message is printed. If an error occurs, the exception message is displayed to help identify the issue.

```
In [1]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt

# Try to connect to the petsdb database
try:
    # Create a connection to the database file
    conn = sqlite3.connect('manoj_project.db')

    # Create a cursor object to execute SQL queries
    cursor = conn.cursor()

    # Print a success message if the connection works
    print("Database connection successful!\n")

# If any error occurs while connecting
except Exception as e:
    # Print the error message
    print("Error connecting to database:\n", e)
```

Database connection successful!

Creating the Final Combined Airport Dataset

In this step, a new table called **final_airport_data** is created by combining information from three different tables: **all_airports** (Dataset from the Wiki), **city_weather** (Dataset from the Weather API), and **city_population** (Dataset from the Kaggle csv file).

First, the code removes the table if it already exists to avoid duplication errors when running the notebook multiple times. Then, a new table is created using a **LEFT JOIN** to merge the datasets based on the common fields **City** and **State**. This ensures that all airport records are kept, even if weather or population data is missing for some cities.

The resulting table contains airport details, along with corresponding **weather information and population data**, making it easier to perform further analysis in a single unified dataset. Finally, the changes are saved to the database using a commit operation.

```
In [2]: # Drop the table if it already exists
cursor.execute("DROP TABLE IF EXISTS final_airport_data")

# Create the final table joining all_airports, city_weather, and city_population
cursor.execute("""
CREATE TABLE final_airport_data AS
SELECT
    a.*,
    w.*,
    p.*
FROM all_airports a
LEFT JOIN city_weather w
    ON a.City = w.City AND a.State = w.State
LEFT JOIN city_population p
    ON a.City = p.City AND a.State = p.State
""")

# Commit the changes
conn.commit()
```

Verify Tables in the SQLite Database

After creating the final dataset, confirm that the tables were successfully created in the SQLite database. The following code queries the database system table (`sqlite_master`) to retrieve the names of all tables currently stored in the database.

This helps verify that the expected tables exist and ensures that the database structure is correct before continuing with further analysis.

This query returns a list of all tables in the connected database.

```
In [3]: # Check tables in the database after creating the final dataset
query = "SELECT name FROM sqlite_master WHERE type='table';"
pd.read_sql(query, conn)
```

Out[3]:

	name
0	city_population
1	large_hub_airports
2	medium_hub_airports
3	busiest_airports
4	all_airports
5	city_weather
6	final_airport_data

Load Data from SQLite Database

In this step, the data from the SQLite database is loaded into Pandas DataFrames for analysis.

Three tables are read from the database:

- **all_airports** – Contains airport traffic and airport details.
- **city_weather** – Contains weather information for the cities where airports are located.
- **city_population** – Contains population information for those cities.

Each table is loaded into a separate Pandas DataFrame using `pd.read_sql()`.

After loading the data, the total number of records in each DataFrame is printed to confirm that the data was successfully retrieved from the database.

```
In [4]: # Read all records from the 'all_airports' table into a pandas DataFrame
airport_df = pd.read_sql("SELECT * FROM all_airports", conn)

# Print the total number of records in the airport DataFrame
print("\nTotal Records in airport_df:", len(airport_df))

# Read all records from the 'city_weather' table into a pandas DataFrame
weather_df = pd.read_sql("SELECT * FROM city_weather", conn)

# Print the total number of records in the weather DataFrame
print("\nTotal Records in weather_df:", len(weather_df))

# Read all records from the 'city_population' table into a pandas DataFrame
population_df = pd.read_sql("SELECT * FROM city_population", conn)

# Print the total number of records in the population DataFrame
print("\nTotal Records in population_df:", len(population_df))
```

Total Records in airport_df: 93

Total Records in weather_df: 59

Total Records in population_df: 28450

Load Final Airport Data from SQLite

The following code reads all records from the `final_airport_data` table in the SQLite database and loads them into a Pandas DataFrame called `final_df`. After loading the data, it prints the DataFrame's total number of rows to confirm how many records were retrieved from the database.

This step helps verify that the data has been successfully loaded and allows us to understand the dataset size before performing further analysis.

```
In [5]: # Read all data from the final_airport_data table into a pandas DataFrame
final_df = pd.read_sql("SELECT * FROM final_airport_data", conn)

# Print the total number of records (rows) in the DataFrame
print("\nTotal Records in final_df:", len(final_df))
```

Total Records in final_df: 94

Displaying Column Names of the DataFrame

The following code is used to display the names of all columns present in the `final_df` DataFrame. This helps verify the dataset's structure and ensures the expected columns are available for further analysis or processing.

By checking the column names, we can confirm that the data was loaded and combined correctly before performing additional operations such as filtering, aggregation, or visualization.

```
In [26]: # Display the column names of the DataFrame
final_df.columns
```

```
Out[26]: Index(['Rank', 'Airport', 'IATA_Code', 'City', 'Metro_Area', 'State',
               'Passengers_2024', 'Passengers_2023', 'Passengers_2022',
               'Passengers_2021', 'Passengers_2020', 'Passengers_2019',
               'Passengers_2018', 'Passengers_2017', 'Passengers_2016',
               'Passengers_2015', 'Pct_Change_24_23', 'Airport_Type', 'City:1',
               'State:1', 'Country', 'Dt', 'Id', 'Name', 'Coord.Lon', 'Coord.Lat',
               'Temperature_F', 'Feels_Like_F', 'Temp_Min_F', 'Temp_Max_F', 'Pressure',
               'Humidity_Pct', 'Sea_Level', 'Grnd_Level', 'Wind_Speed_Mph',
               'Wind_Direction_Deg', 'Clouds.All', 'Sys.Type', 'Sys.Id',
               'Country_Code', 'Sys.Sunrise', 'Sys.Sunset', 'Gust', 'Snow.1H',
               'Rain.1H', 'Weather_Id', 'Weather_Main', 'Weather_Description',
               'Weather_Icon', 'Timestamp', 'Sunrise', 'Sunset', 'Temp_Range_F',
               'Is_Daytime', 'INDEX', 'TABLE_ID', 'CITY:2', 'SUMMARY_LEVEL',
               'PLACE_FIPS', 'GEO_ID', 'STATE:2', 'STATE_FIPS', 'TOTAL_POPULATION'],
              dtype='object')
```


Visualization 1 – Top 10 Airports by Passenger Traffic (2024)

This code sorts the dataset based on the number of passengers in **2024** and selects the **top 10 airports with the highest passenger traffic**.

A **bar chart** is then created using Matplotlib to visually compare the passenger counts of these airports.

- The dataframe is sorted in **descending order** using the `Passengers_2024` column.
- The **top 10 airports** are selected using `.head(10)`.
- A **bar chart** is created where:
 - The **x-axis** represents the airport names.
 - The **y-axis** represents the number of passengers in 2024.
- Airport labels are **rotated for better readability**.
- The chart includes a **title and axis labels** to clearly describe the visualization.

This visualization helps identify the **busiest airports based on passenger traffic in 2024**, making it easier to analyze airport activity and compare passenger volumes across major airports.

```
In [19]: # Select top 10 airports by Passengers_2024 with distinct airport names
top_airports = (
    final_df.drop_duplicates(subset="Airport") # remove duplicate airport names
    .sort_values("Passengers_2024", ascending=False)
    .head(10)
)

# Create a new column for unique labels by combining Airport name and IATA code
top_airports["Airport_Label"] = top_airports["Airport"] + " (" + top_airports["IATA"]

# Create a new figure for the bar chart
plt.figure(figsize=(12,6))

# Define a list of 10 colors, one for each airport
colors = [
    "red", "blue", "green", "orange", "purple",
    "cyan", "magenta", "yellow", "brown", "pink"
]

# Create a bar chart with different colors for each airport
plt.bar(top_airports["Airport_Label"], top_airports["Passengers_2024"], color=colors)

# Rotate the airport names on the x-axis for better readability
plt.xticks(rotation=45, ha='right')

# Add a title to the chart
plt.title("Top 10 Airports by Passenger Traffic (2024)")

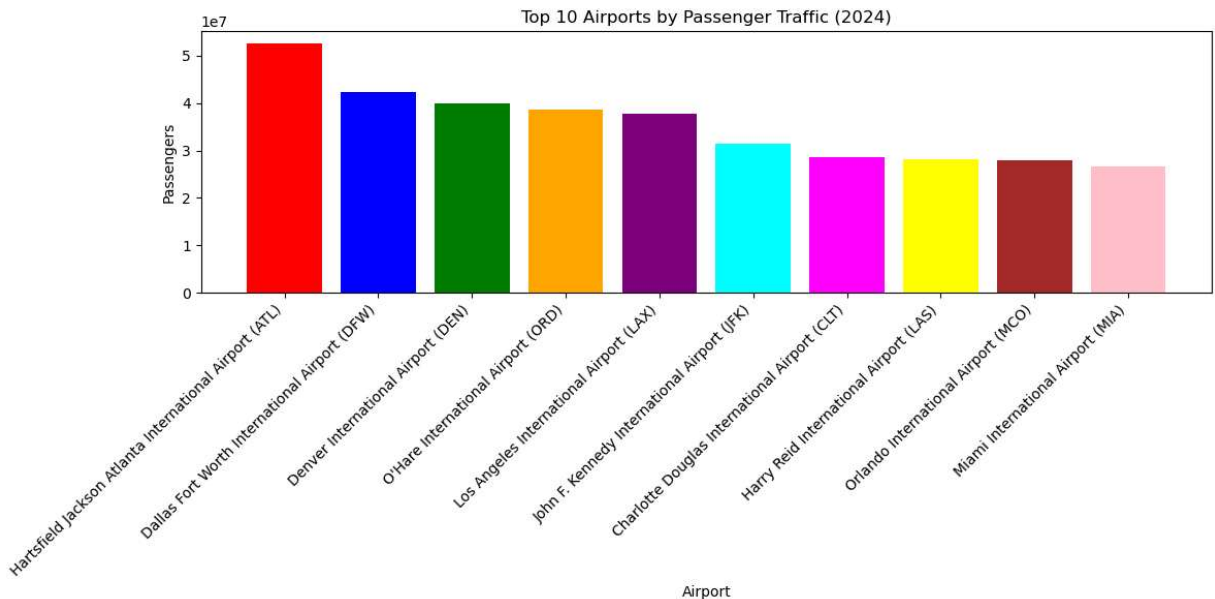
# Label the x-axis
plt.xlabel("Airport")

# Label the y-axis
```

```
plt.ylabel("Passengers")

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the chart
plt.show()
```



Visualization 2 – Airport Type Distribution

The following code analyzes the distribution of airport categories in the dataset and visualizes the results using a bar chart.

First, the `value_counts()` function is used to count the number of airports in each `Airport_Type` category in the `final_df` DataFrame. This helps summarize the number of airports by type.

Next, a new plot figure is created using Matplotlib. The airport type counts are then displayed as a **bar chart**, which makes it easy to compare the number of airports in each category visually.

The chart is improved by adding a **title** and labeling both the **x-axis** (Airport Type) and **y-axis** (Count). Finally, `plt.show()` is used to display the chart.

This visualization helps quickly understand how airports are distributed across different airport types in the dataset.

```
In [24]: # Count the number of airports for each type
airport_type_counts = final_df["Airport_Type"].value_counts()

# Create a new figure for the plot
plt.figure(figsize=(8,5))

# Define colors for each airport type
```

```

# Adjust the colors as needed for your dataset
colors = {
    "Large_Hub": "red",
    "Medium_Hub": "blue",
    "Busiest": "green"
}

# Map the colors to the order of the bars
bar_colors = [colors[atype] for atype in airport_type_counts.index]

# Plot the counts as a bar chart with specified colors
airport_type_counts.plot(kind="bar", color=bar_colors)

# Add a title to the chart
plt.title("Distribution of Airport Types")

# Label the x-axis
plt.xlabel("Airport Type")

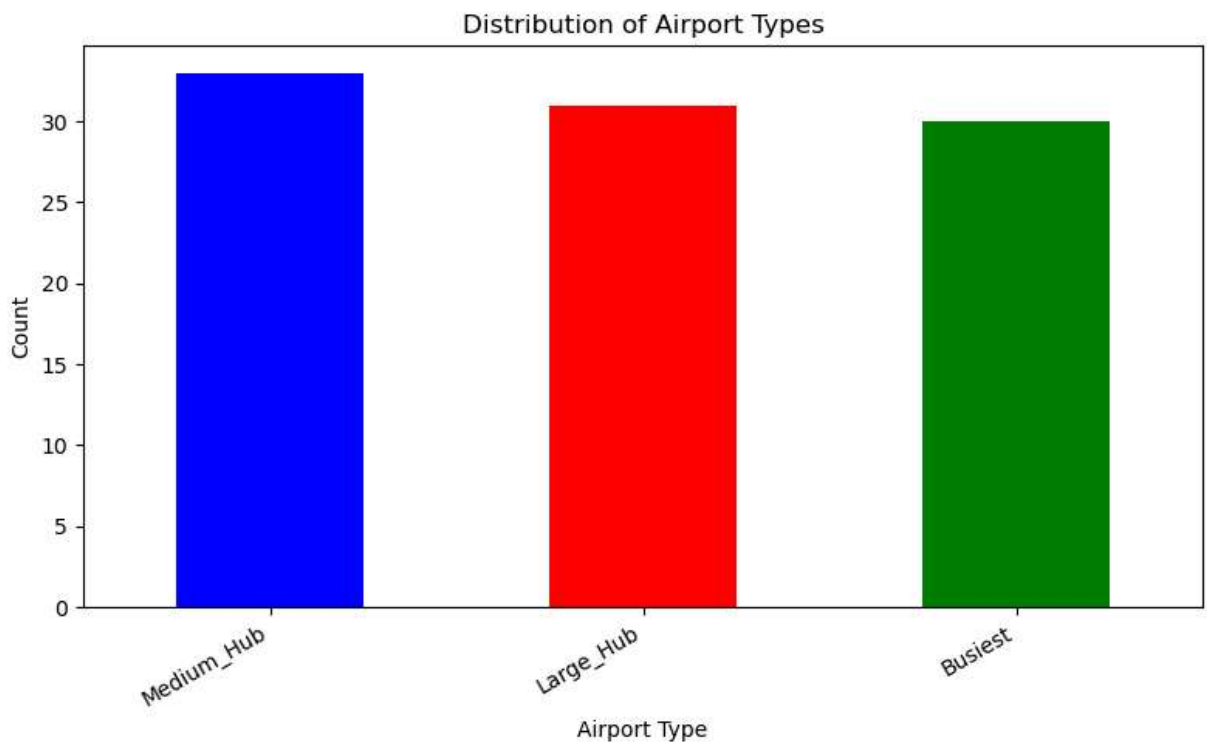
# Label the y-axis
plt.ylabel("Count")

# Rotate x-axis labels slightly for readability
plt.xticks(rotation=30, ha='right')

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the plot
plt.show()

```



Visualization 3 – Population vs Passenger Traffic

The following code creates a scatter plot to visualize the relationship between the **total population of a city** and the **number of airport passengers in 2024**.

- A new plot figure is created using Matplotlib.
- A scatter plot is generated using the `TOTAL_POPULATION` column for the x-axis and `Passengers_2024` for the y-axis.
- The chart helps analyze whether cities with larger populations tend to have higher airport passenger traffic.
- The plot also includes a title and labels for both axes to make the visualization easier to understand.

Finally, the plot is displayed using `plt.show()`.

```
In [63]: # Create a new figure for the plot
plt.figure()

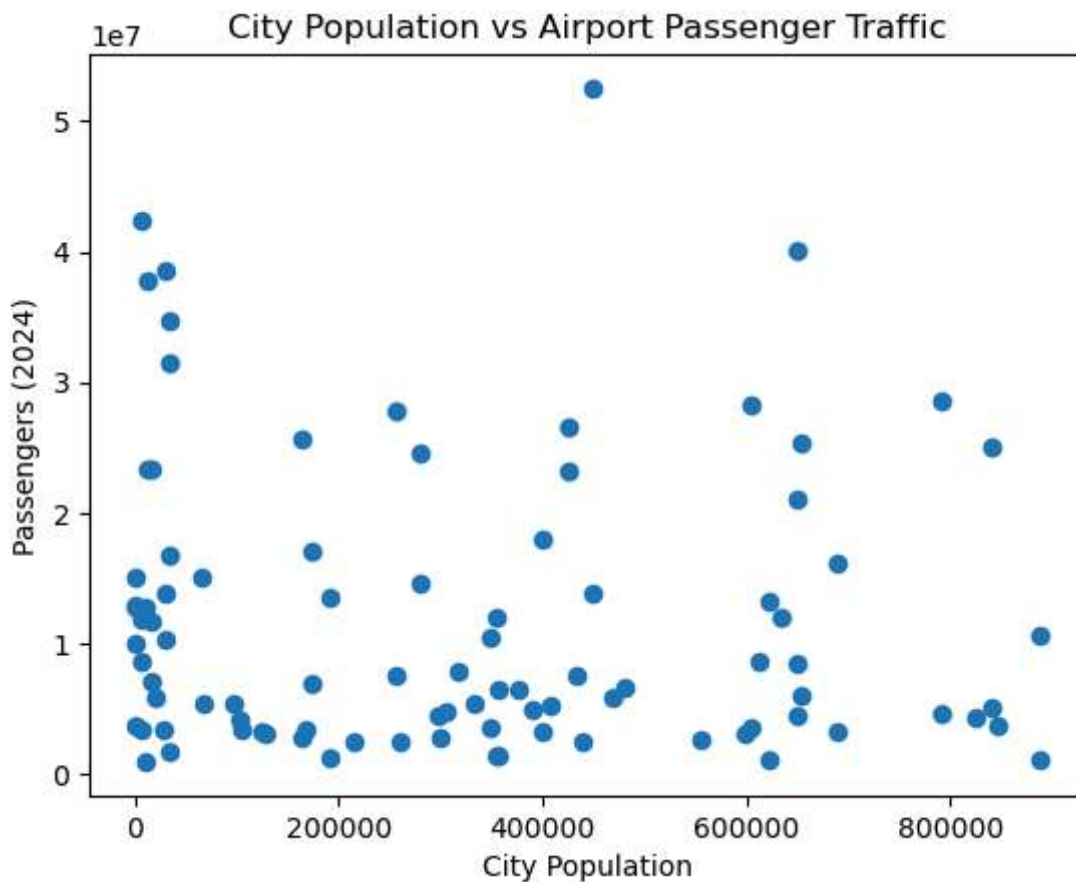
# Create a scatter plot of city population vs. airport passengers in 2024
plt.scatter(final_df["TOTAL_POPULATION"], final_df["Passengers_2024"])

# Add a title to the plot
plt.title("City Population vs Airport Passenger Traffic")

# Label the x-axis
plt.xlabel("City Population")

# Label the y-axis
plt.ylabel("Passengers (2024)")

# Display the plot
plt.show()
```



Visualization 4 – Scatter Plot: Temperature vs Passenger Traffic

The following code creates a scatter plot to visualize the relationship between the **temperature** of a city and the **number of airport passengers in 2024**.

- A new plot figure is created using Matplotlib.
- A scatter plot is generated using the `Temperature_F` column for the x-axis and `Passengers_2024` for the y-axis.
- The chart helps analyze whether cities with higher or lower temperatures tend to have more airport passenger traffic.
- The plot also includes a title and labels for both axes to make the visualization easier to understand.

Finally, the plot is displayed using `plt.show()`.

```
In [27]: # Create a new figure for the plot
plt.figure()

# Create a scatter plot with Temperature on X-axis and Passengers on Y-axis
plt.scatter(final_df["Temperature_F"], final_df["Passengers_2024"])

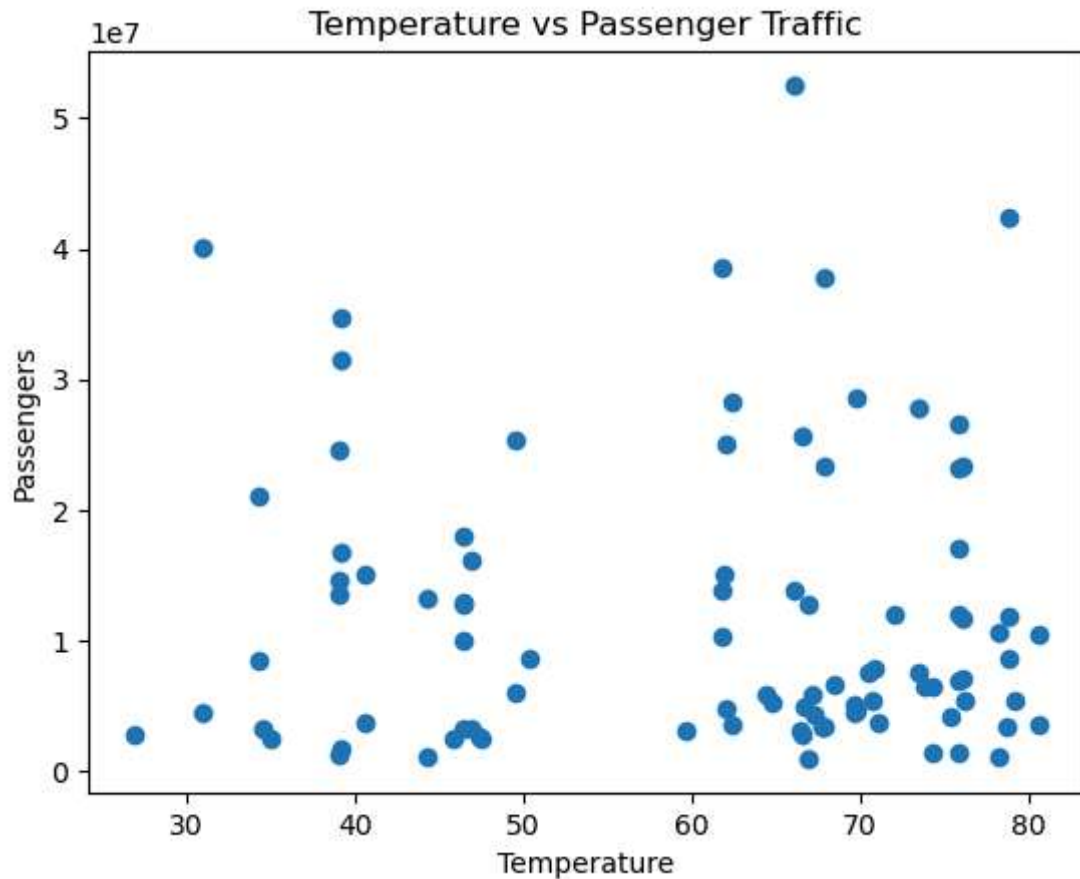
# Add a title to the plot
plt.title("Temperature vs Passenger Traffic")

# Label the X-axis
```

```
plt.xlabel("Temperature")

# Label the Y-axis
plt.ylabel("Passengers")

# Display the plot
plt.show()
```



Visualization 5 – Line Chart: Passenger Trend Over Years

The following code creates a line chart to visualize the **trend of total airport passengers from 2018 to 2024**.

- A list of columns representing passenger counts for each year is defined.
- The total passengers for each year are calculated by summing the values across all cities.
- A new plot figure is created using Matplotlib.
- The passenger trend is plotted as a **line chart with markers** to highlight each year's data point.
- The chart includes a title and labels for both axes to make the trend easy to interpret.

Finally, the plot is displayed using `plt.show()`.

```
In [28]: # List of columns representing passenger counts for each year
years = ["Passengers_2018", "Passengers_2019", "Passengers_2020",
        "Passengers_2021", "Passengers_2022", "Passengers_2023", "Passengers_2024"]
```

```

# Calculate the total passengers for each year
trend = final_df[years].sum()

# Create a new figure for the plot
plt.figure(figsize=(10,6)) # Optional: make figure larger for readability

# Plot the passenger trend as a Line chart with markers for each year
trend.plot(kind="line", marker="o")

# Set the title of the plot
plt.title("Passenger Trend Over Years")

# Label the x-axis
plt.xlabel("Year")

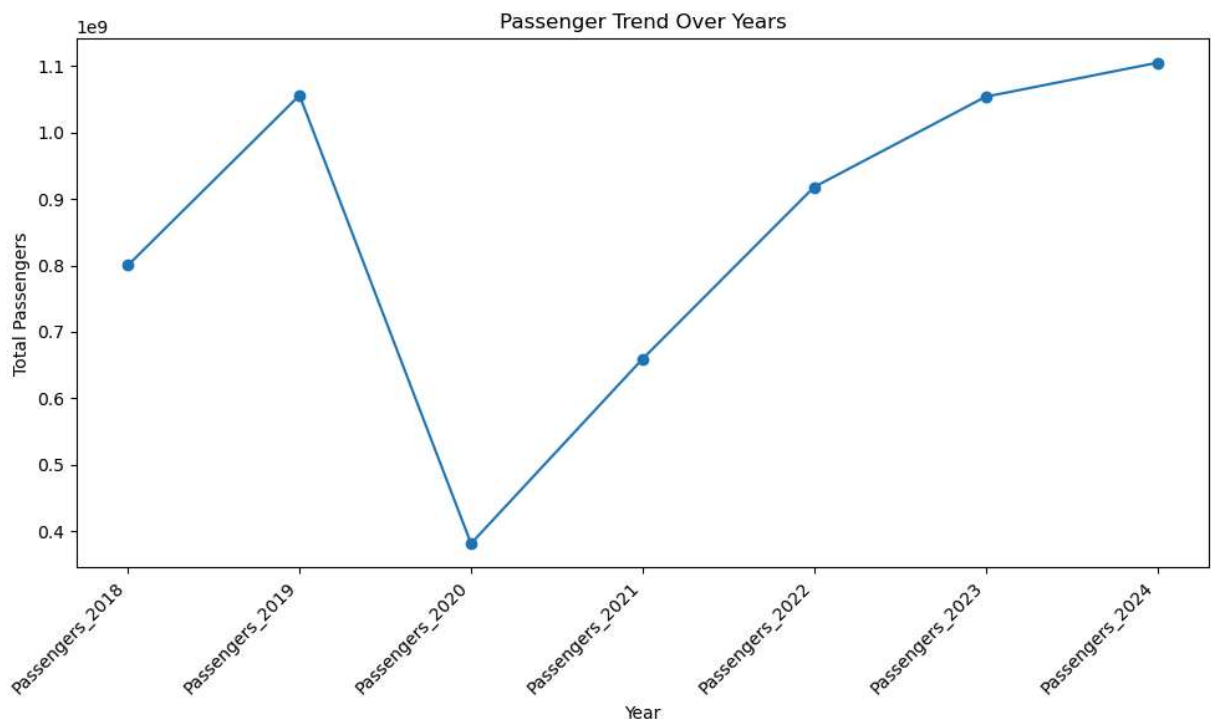
# Label the y-axis
plt.ylabel("Total Passengers")

# Rotate x-axis labels for clarity
plt.xticks(rotation=45, ha='right')

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the plot
plt.show()

```



Visualization 6 — Scatter Plot: Airport Passenger Traffic (2019) vs Airport Passenger Traffic (2024)

The following code creates a scatter plot to visualize the relationship between the **number of airport passengers in 2019** and the **number of airport passengers in 2024**.

- A new plot figure is created using Matplotlib.
- A scatter plot is generated using the `Passengers_2019` column for the x-axis and `Passengers_2024` for the y-axis.
- The plot also includes a title and labels for both axes to make the visualization easier to understand.

Finally, the plot is displayed using `plt.show()`.

```
In [29]: # Create a new figure for the plot
plt.figure()

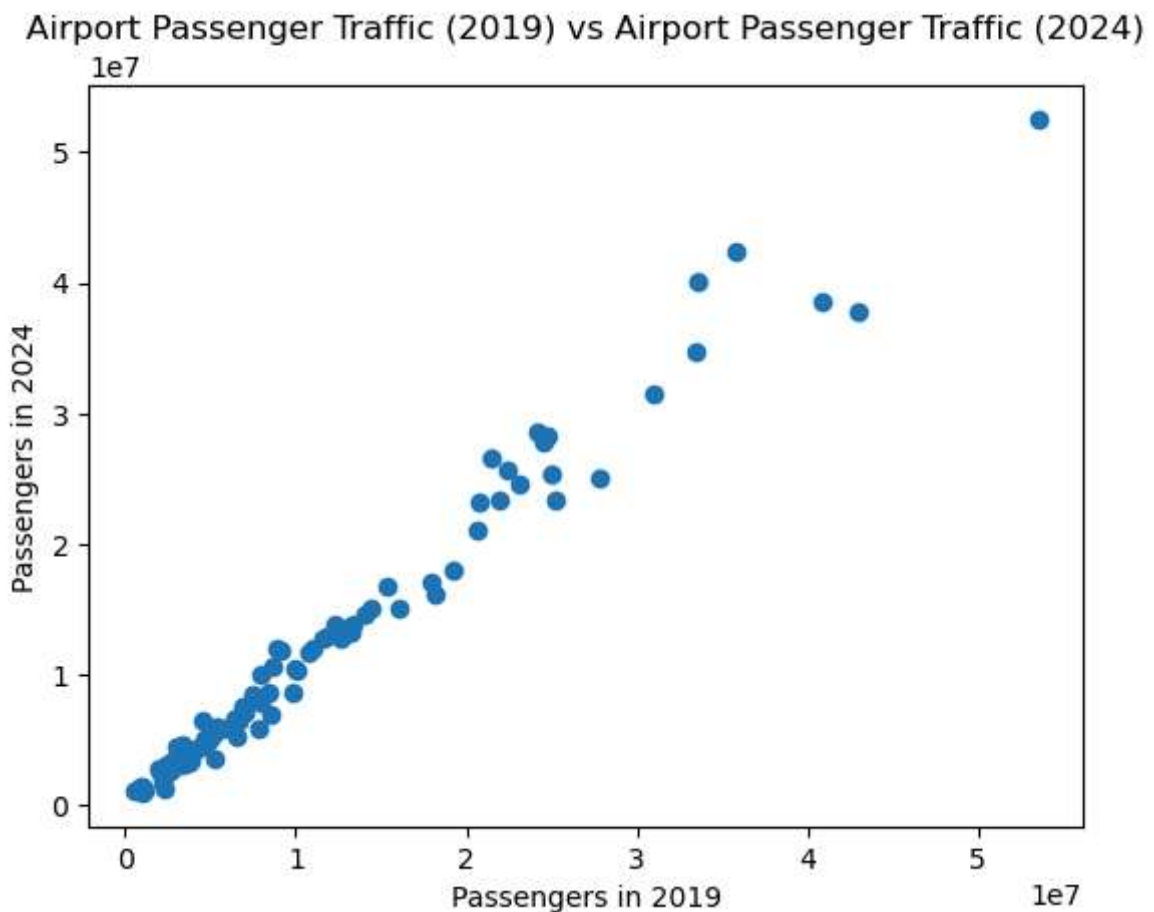
# Create a scatter plot with 2019 passengers on the x-axis and the 2024 passengers
plt.scatter(final_df["Passengers_2019"], final_df["Passengers_2024"])

# Set the title of the plot
plt.title("Airport Passenger Traffic (2019) vs Airport Passenger Traffic (2024)")

# Label the x-axis
plt.xlabel("Passengers in 2019")

# Label the y-axis
plt.ylabel("Passengers in 2024")

# Display the plot
plt.show()
```



Visualization 7 — Bar Chart: Top 10 Airports by Passenger Growth (2022–2024)

The following code calculates the **passenger growth** for each airport between 2022 and 2024 and visualizes the **top 10 airports** with the highest growth.

- A new column `Passenger_Growth` is added to `final_df` by subtracting `Passengers_2022` from `Passengers_2024`.
- The top 10 airports with the largest growth are selected by sorting the data in descending order.
- A bar chart is created to show the **airport codes** on the x-axis and their corresponding **passenger growth** on the y-axis.
- The chart includes a title and axis labels for clarity.
- Finally, the bar chart is displayed using `plt.show()`.

```
In [37]: # Calculate the passenger growth between 2022 and 2024
final_df["Passenger_Growth"] = final_df["Passengers_2024"] - final_df["Passengers_2022"]

# Get the top 10 airports with the highest passenger growth
top_growth = (
    final_df.drop_duplicates(subset="Airport") # remove duplicate airport names
    .sort_values("Passenger_Growth", ascending=False)
    .head(10)
)

# Create a new figure for the bar chart
plt.figure(figsize=(12,6))

# Define 10 distinct colors, one for each airport
colors = [
    "red", "blue", "green", "orange", "purple",
    "cyan", "magenta", "yellow", "brown", "pink"
]

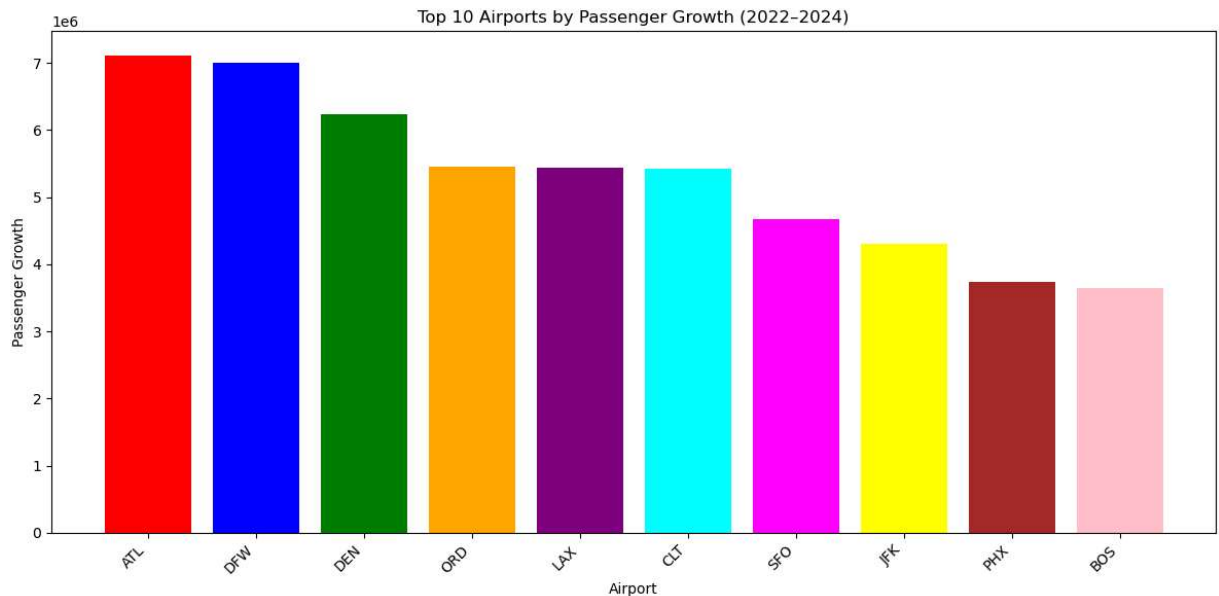
# Plot the bar chart with unique labels and colors
plt.bar(top_growth["IATA_Code"], top_growth["Passenger_Growth"], color=colors)

# Rotate the x-axis labels for readability
plt.xticks(rotation=45, ha='right')

# Add a title and axis labels
plt.title("Top 10 Airports by Passenger Growth (2022–2024)")
plt.xlabel("Airport")
plt.ylabel("Passenger Growth")

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the chart
plt.show()
```



Visualization 8 — Bar Chart: Average City Population by Airport Type

The following code calculates and visualizes the **average number of passengers in 2024** for each **airport type**.

- First, the data is grouped by the `Airport_Type` column, and the mean of `Passengers_2024` is calculated.
- A new figure is created using Matplotlib.
- The average passengers for each airport type are displayed as a **bar chart**.
- The chart includes a title and labels for both axes to clearly indicate the airport type and average passenger traffic.
- This visualization helps compare airport traffic across different types of airports.

Finally, the bar chart is displayed using `plt.show()`.

```
In [41]: # Calculate the average number of passengers in 2024 for each airport type
avg_population = final_df.groupby("Airport_Type")["Passengers_2024"].mean()

# Create a new figure for the plot
plt.figure(figsize=(8,5))

# Define colors for each airport type (adjust according to your dataset)
colors = {
    "Large_Hub": "red",
    "Medium_Hub": "blue",
    "Busiest": "green"
}

# Map the colors to the order of the bars
bar_colors = [colors[atype] for atype in avg_population.index]

# Plot the average passengers as a bar chart with specified colors
avg_population.plot(kind="bar", color=bar_colors)
```

```
# Set the title of the chart
plt.title("Average City Population by Airport Type")

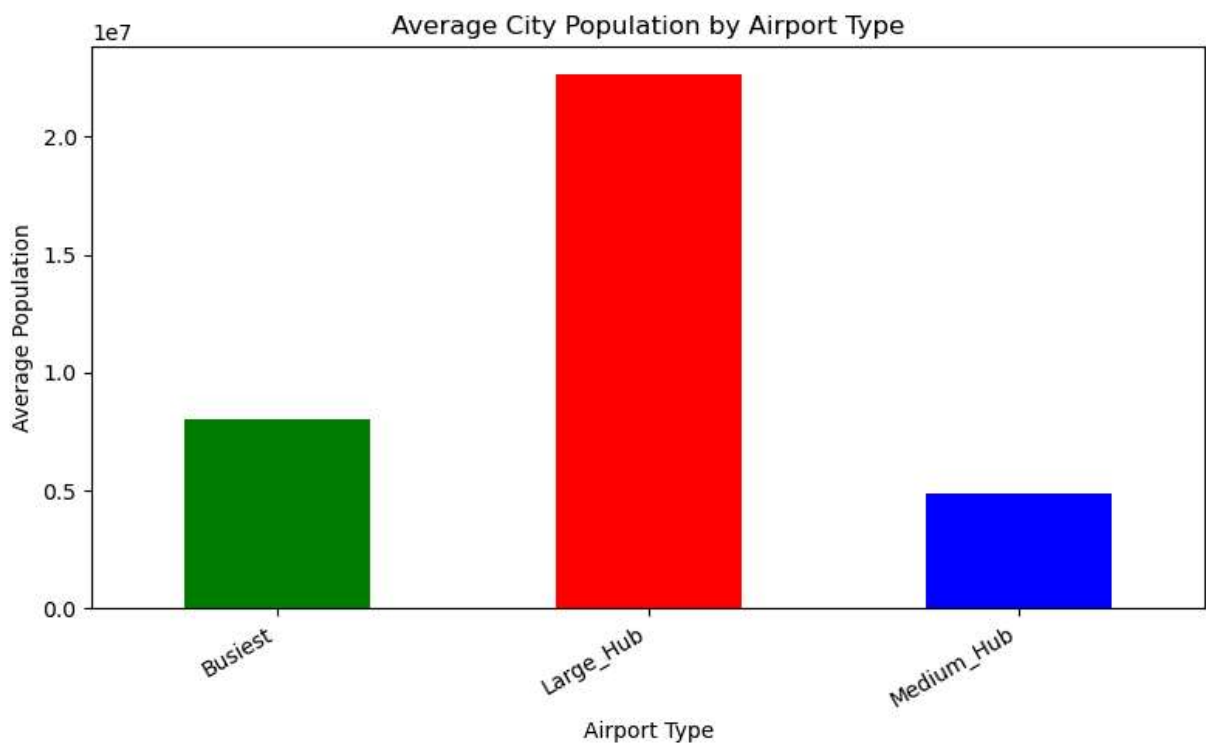
# Label the x-axis
plt.xlabel("Airport Type")

# Label the y-axis
plt.ylabel("Average Population")

# Rotate x-axis labels for better readability
plt.xticks(rotation=30, ha='right')

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the plot
plt.show()
```



Visualization 9 — Scatter Plot: Weather Temperature vs Airport

The following code creates a scatter plot to visualize the relationship between a city's **Weather temperature** and the **airport**.

- A new plot figure is created using Matplotlib.
- A scatter plot is generated using the `Temperature_F` column for the x-axis and `IATA_Code` for the y-axis.
- The chart helps analyze which cities' airports have higher or lower temperatures.
- The plot also includes a title and labels for both axes to make the visualization easier to interpret.

Finally, the plot is displayed using `plt.show()`.

```
In [44]: # Create a new figure with larger size
plt.figure(figsize=(10,12))

# Create a horizontal scatter plot: Passengers vs Airport
plt.scatter(final_df["Temperature_F"], final_df["IATA_Code"], color='blue')

# Add a title to the plot
plt.title("Weather Temperature vs Airport")

# Label the x-axis
plt.xlabel("Weather Temperature (°F)")

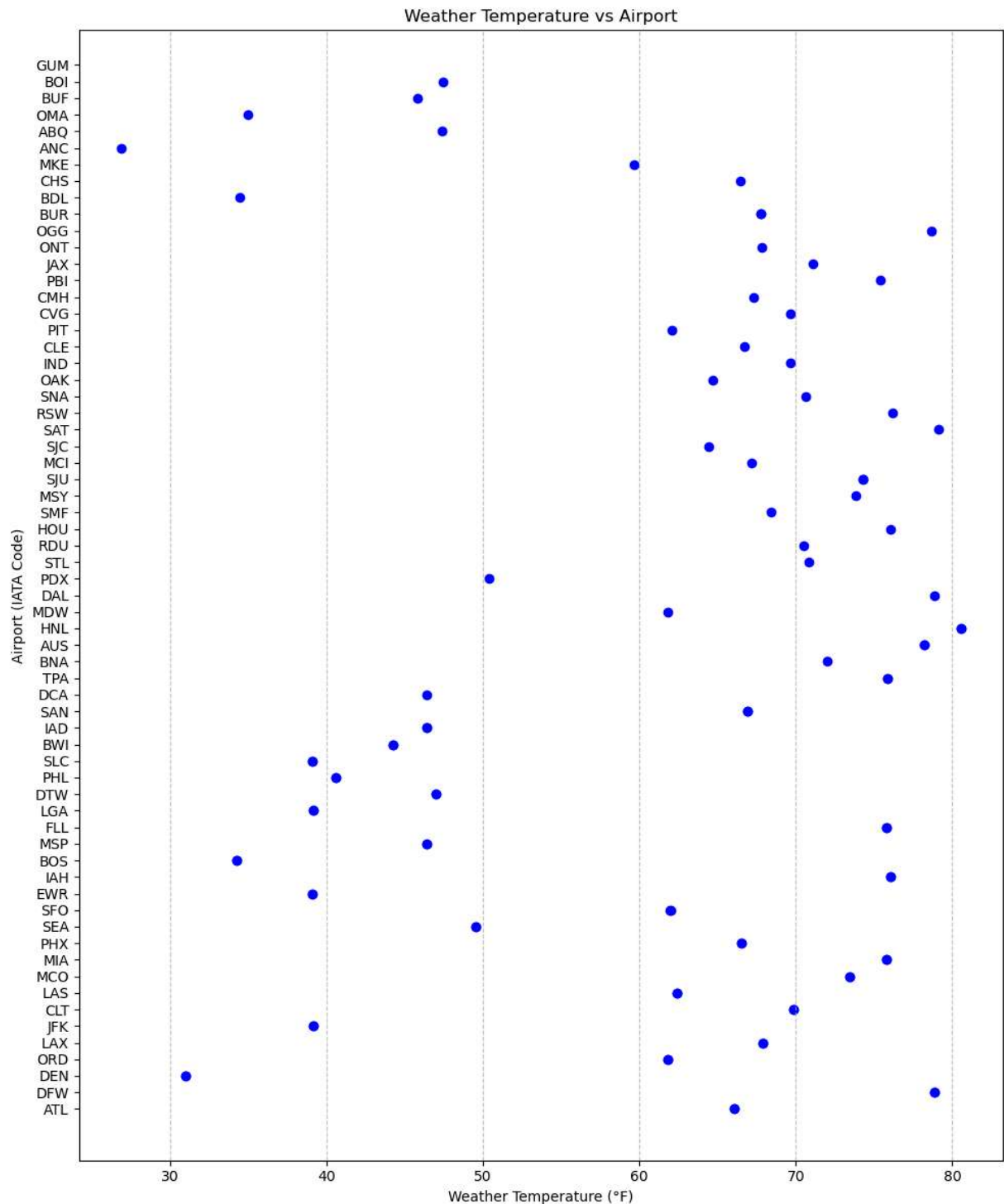
# Label the y-axis
plt.ylabel("Airport (IATA Code)")

# Rotate y-axis labels slightly for readability
plt.yticks(rotation=0) # 0° keeps them horizontal, better for horizontal scatter

# Add gridlines for easier reading
plt.grid(axis='x', linestyle='--', alpha=0.7)

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the plot
plt.show()
```



Project Summary and Ethical Considerations

This project involved cleaning, transforming, storing, and analyzing airport data from three different datasets: airport traffic data (**all_airports**), weather data (**city_weather**), and population data (**city_population**). The datasets were first cleaned and standardized so that they could be joined using common fields such as **City** and **State**. After cleaning, each dataset was loaded into a **SQLite database** as individual tables. SQL joins were then used to merge these tables into a consolidated dataset called **final_airport_data**. This allowed information from multiple sources to be analyzed together.

Several transformations were performed on the data. Missing values were handled, column names were standardized, and a new column, **Airport_Type**, was created to categorize airports by passenger count. Additionally, the three datasets were merged using SQL joins to produce a single dataset suitable for visualization and analysis.

There are important ethical considerations when performing data cleansing and transformation. One risk is that modifying or filtering data may unintentionally introduce bias or misrepresent the original dataset. For example, if missing values are incorrectly replaced or assumptions are made about airport classifications, the resulting analysis may be misleading. Another concern involves data accuracy and credibility. The datasets used in this project were sourced from publicly available sources and were deemed credible. However, it is always important to verify the data's origin and ensure it complies with any applicable legal or regulatory guidelines.

During the cleaning process, assumptions were made about how to interpret missing or zero values. For instance, airports with zero passenger traffic in certain years were categorized as **"Busiest"** in the dataset. Such assumptions can introduce risk if they do not accurately represent real-world conditions. But this is due to data unavailability.

To mitigate ethical concerns, it is important to clearly document all transformations, assumptions, and data sources. Transparency ensures that others can understand how the dataset was prepared and evaluate whether the conclusions drawn from the data are valid. Ensuring data is obtained from reputable sources and maintaining proper documentation helps promote ethical data analysis practices.

Final Dataset Output (Human Readable)

Display Airport Dataset

The following code displays the **entire Airport dataset** stored in the `airport_df` DataFrame.

- This allows you to **view all records** in the dataset.
- Useful for quickly checking the **structure, column names, and data values**.
- Helps in understanding the dataset before performing any **analysis or visualization**.

Simply running `airport_df` prints all rows and columns in the notebook output.

```
In [70]: # Print the Airport dataset to display all the records
airport_df
```


Out[70]:

	Rank	Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
0	1	Hartsfield Jackson Atlanta International Airport	ATL	Atlanta	Metro Atlanta	GA	52511402
1	2	Dallas Fort Worth International Airport	DFW	Dallas	Dallas-Fort Worth	TX	42351316
2	3	Denver International Airport	DEN	Denver	Greater Denver	CO	40012895
3	4	O'Hare International Airport	ORD	Chicago	Chicago	IL	38575693
4	5	Los Angeles International Airport	LAX	Los Angeles	Greater Los Angeles	CA	37760834
5	6	John F. Kennedy International Airport	JFK	New York City	New York Metro	NY	31466102
6	7	Charlotte Douglas International Airport	CLT	Charlotte	Greater Charlotte	NC	28523822
7	8	Harry Reid International Airport	LAS	Las Vegas	Las Vegas Valley	NV	28244966
8	9	Orlando International Airport	MCO	Orlando	Greater Orlando	FL	27859783
9	10	Miami International Airport	MIA	Miami	Miami Metro	FL	26588002
10	11	Phoenix Sky Harbor International Airport	PHX	Phoenix	Metro Phoenix	AZ	25595723
11	12	Seattle Tacoma International Airport	SEA	Seattle	Seattle Metro	WA	25414592
12	13	San Francisco International Airport	SFO	San Francisco	San Francisco Bay Area	CA	25078968

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
13	14	Newark Liberty International Airport	EWR	Newark	New York Metro	NJ	24544320
14	15	George Bush Intercontinental Airport	IAH	Houston	Greater Houston	TX	23349157
15	16	Logan International Airport	BOS	Boston	Greater Boston	MA	21090721
16	17	Minneapolis Saint Paul International Airport	MSP	Minneapolis	Minneapolis–Saint Paul	MN	18054481
17	18	Fort Lauderdale Hollywood International Airport	FLL	Fort Lauderdale	Miami Metro	FL	17096131
18	19	Laguardia Airport	LGA	New York City	New York Metro	NY	16715567
19	20	Detroit Metropolitan Airport	DTW	Detroit	Detroit Metro	MI	16110696
20	21	Philadelphia International Airport	PHL	Philadelphia	Philadelphia Metro	PA	15102261
21	22	Salt Lake City International Airport	SLC	Salt Lake City	Wasatch Front	UT	13543570
22	23	Baltimore Washington International Airport	BWI	Baltimore	Baltimore metropolitan area	MD	13221461
23	24	Dulles International Airport	IAD	Washington, D.C.	Washington Metro	VA	13003234
24	25	San Diego International Airport	SAN	San Diego	Greater San Diego	CA	12780013
25	26	Ronald Reagan Washington National Airport	DCA	Washington, D.C.	Washington Metro	VA	12750892

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
26	27	Tampa International Airport	TPA	Tampa	Tampa Bay area	FL	12075591
27	28	Nashville International Airport	BNA	Nashville	Greater Nashville	TN	12058688
28	29	Austin Bergstrom International Airport	AUS	Austin	Greater Austin	TX	10678073
29	30	Daniel K. Inouye International Airport	HNL	Honolulu	Oahu	HI	10449022
30	31	Midway International Airport	MDW	Chicago	Chicago	IL	10360093
31	32	Dallas Love Field	DAL	Dallas	Dallas–Fort Worth	TX	8654991
32	33	Portland International Airport	PDX	Portland	Portland	OR	8639088
33	34	St. Louis Lambert International Airport	STL	St. Louis	St. Louis	MO	7807362
34	35	Raleigh Durham International Airport	RDU	Raleigh	Research Triangle	NC	7584394
35	36	William P. Hobby Airport	HOU	Houston	Houston	TX	7116967
36	37	Sacramento International Airport	SMF	Sacramento	Sacramento	CA	6679426
37	38	Louis Armstrong New Orleans International Airport	MSY	New Orleans	New Orleans	LA	6537092
38	39	Luis Muñoz Marín International Airport	SJU	San Juan	San Juan	PR	6490669

	Rank	Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
39	40	Kansas City International Airport	MCI	Kansas City	Kansas City	MO	5915078
40	41	Norman Y. Mineta San José International Airport	SJC	San Jose	San Francisco Bay Area	CA	5822019
41	42	San Antonio International Airport	SAT	San Antonio	San Antonio	TX	5466684
42	43	Southwest Florida International Airport	RSW	Fort Myers	Southwest Florida	FL	5425293
43	44	John Wayne Airport	SNA	Santa Ana	Greater Los Angeles	CA	5370273
44	45	Oakland San Francisco Bay Airport	OAK	Oakland	San Francisco Bay Area	CA	5292736
45	46	Indianapolis International Airport	IND	Indianapolis	Indianapolis	IN	5181947
46	47	Cleveland Hopkins International Airport	CLE	Cleveland	Cleveland	OH	4950345
47	48	Pittsburgh International Airport	PIT	Pittsburgh	Pittsburgh	PA	4862376
48	49	Cincinnati Northern Kentucky International Air...	CVG	Cincinnati	Cincinnati	OH	4500414
49	50	John Glenn Columbus International Airport	CMH	Columbus	Columbus	OH	4387395
50	51	Palm Beach International Airport	PBI	West Palm Beach	Miami Metro	FL	4127211

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
51	52	Jacksonville International Airport	JAX	Jacksonville	Jacksonville	FL	3749791
52	53	Ontario International Airport	ONT	Ontario	Greater Los Angeles	CA	3494554
53	54	Kahului Airport	OGG	Kahului	Maui	HI	3415615
54	55	Hollywood Burbank Airport	BUR	Burbank	Greater Los Angeles	CA	3383237
55	56	Bradley International Airport	BDL	Hartford	Hartford	CT	3285194
56	57	Charleston International Airport	CHS	Charleston	Charleston	SC	3115194
57	58	Milwaukee Mitchell International Airport	MKE	Milwaukee	Milwaukee	WI	3104054
58	59	Ted Stevens Anchorage International Airport	ANC	Anchorage	Anchorage	AK	2767856
59	60	Albuquerque International Sunport	ABQ	Albuquerque	Albuquerque	NM	2690218
60	61	Eppley Airfield	OMA	Omaha	Omaha	NE	2583212
61	62	Buffalo Niagara International Airport	BUF	Buffalo	Buffalo–Niagara Falls	NY	2499295
62	63	Boise Airport	BOI	Boise	Boise	ID	2475370
63	1	John F. Kennedy International Airport	JFK	New York City	Queens, New York	NY	34765387
64	2	Los Angeles International Airport	LAX	Los Angeles	Los Angeles, California	CA	23438418
65	3	Miami International Airport	MIA	Miami	Miami, Florida	FL	23202731

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
66	4	San Francisco International Airport	SFO	South San Francisco	South San Francisco, California	CA	15083931
67	5	Newark Liberty International Airport	EWR	Newark	Newark, New Jersey	NJ	14690370
68	6	O'Hare International Airport	ORD	Chicago	Chicago, Illinois	IL	13865434
69	7	Hartsfield Jackson Atlanta International Airport	ATL	Atlanta	College Park, Georgia	GA	13813088
70	8	Dallas Fort Worth International Airport	DFW	Dallas	Grapevine, Texas	TX	11841330
71	9	George Bush Intercontinental Airport	IAH	Houston	Houston, Texas	TX	11675134
72	10	Washington Dulles International Airport	IAD	Washington, D.C.	Dulles, Virginia	VA	10060851
73	11	General Edward Lawrence Logan International Airport	BOS	Boston	Boston, Massachusetts	MA	8491518
74	12	Orlando International Airport	MCO	Orlando	Orlando, Florida	FL	7610776
75	13	Fort Lauderdale Hollywood International Airport	FLL	Fort Lauderdale	Fort Lauderdale, Florida	FL	6897993
76	14	Seattle Tacoma International Airport	SEA	Seattle	Seatac, Washington	WA	6056779
77	15	Charlotte Douglas International Airport	CLT	Charlotte	Charlotte, North Carolina	NC	4623143

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
78	16	Denver International Airport	DEN	Denver	Denver, Colorado	CO	4444671
79	17	Philadelphia International Airport	PHL	Philadelphia	Philadelphia, Pennsylvania	PA	3704176
80	18	Harry Reid International Airport	LAS	Las Vegas	Paradise, Nevada	NV	3592102
81	19	Daniel K. Inouye International Airport	HNL	Honolulu	Honolulu, Hawaii	HI	3533266
82	20	Detroit Metropolitan Wayne County Airport	DTW	Detroit	Romulus, Michigan	MI	3353501
83	21	Minneapolis Saint Paul International Airport	MSP	Minneapolis	Saint Paul, Minnesota	MN	3320850
84	22	Phoenix Sky Harbor International Airport	PHX	Phoenix	Phoenix, Arizona	AZ	2821736
85	23	Laguardia Airport	LGA	New York City	Queens, New York	NY	1686395
86	24	Luis Muñoz Marín International Airport	SJU	San Juan	San Juan, Puerto Rico	PR	1458172
87	25	Tampa International Airport	TPA	Tampa	Tampa, Florida	FL	1384192
88	26	Antonio B. Won Pat International Airport	GUM	Tamuning	Tamuning, Guam	Guam	1342793
89	27	Salt Lake City International Airport	SLC	Salt Lake City	Salt Lake City, Utah	UT	1311617

	Rank	Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
90	28	Baltimore Washington International Airport	BWI	Baltimore	Linthicum Heights, Maryland	MD	1156882
91	29	Austin Bergstrom International Airport	AUS	Austin	Austin, Texas	TX	1064668
92	30	San Diego International Airport	SAN	San Diego	San Diego, California	CA	995537

Display Weather Dataset

The following code displays the **entire Weather dataset** stored in the `weather_df` DataFrame.

- Printing the DataFrame allows us to **inspect all records** and check for missing or inconsistent data.
- This step is useful for **understanding the structure** of the dataset, including columns such as temperature, humidity, and weather conditions.
- Viewing the dataset helps in planning **data cleaning, transformation, and analysis** steps.

```
In [71]: # Print the Weather dataset to display all the records
weather_df
```


Out[71]:

	City	State	Country	Dt	Id	Name	Coord.Lon	Coord.Lat
0	Atlanta	GA	US	1772847912	4180439	Atlanta	-84.3880	33.7490
1	Dallas	TX	US	1772848070	4684904	Dallas	-96.7836	32.7668
2	Denver	CO	US	1772848183	5419384	Denver	-104.9847	39.7392
3	Chicago	IL	US	1772848056	4887398	Chicago	-87.6500	41.8500
4	Los Angeles	CA	US	1772847969	5368361	Los Angeles	-118.2437	34.0522
5	New York City	NY	US	1772848036	5128581	New York	-74.0060	40.7143
6	Charlotte	NC	US	1772847965	4460243	Charlotte	-80.8431	35.2271
7	Las Vegas	NV	US	1772847945	5506956	Las Vegas	-115.1372	36.1750
8	Orlando	FL	US	1772848055	4167147	Orlando	-81.3792	28.5383
9	Miami	FL	US	1772847723	4164138	Miami	-80.1937	25.7743
10	Phoenix	AZ	US	1772848188	5308655	Phoenix	-112.0740	33.4484
11	Seattle	WA	US	1772847891	5809844	Seattle	-122.3321	47.6062
12	San Francisco	CA	US	1772847900	5391959	San Francisco	-122.4194	37.7749
13	Newark	NJ	US	1772848137	5101798	Newark	-74.1724	40.7357

	City	State	Country	Dt	Id	Name	Coord.Lon	Coord.Lat
14	Houston	TX	US	1772847846	4699066	Houston	-95.3633	29.7633
15	Boston	MA	US	1772848169	4930956	Boston	-71.0598	42.3584
16	Minneapolis	MN	US	1772848115	5037649	Minneapolis	-93.2638	44.9800
17	Fort Lauderdale	FL	US	1772848115	4155966	Fort Lauderdale	-80.1434	26.1223
18	Detroit	MI	US	1772847930	4990729	Detroit	-83.0458	42.3314
19	Philadelphia	PA	US	1772848059	4560349	Philadelphia	-75.1638	39.9523
20	Salt Lake City	UT	US	1772848195	5780993	Salt Lake City	-111.8911	40.7608
21	Baltimore	MD	US	1772848173	4347778	Baltimore	-76.6122	39.2904
22	Washington, D.C.	VA	US	1772848196	4361885	Washington DC.	-77.0366	38.8950
23	San Diego	CA	US	1772848197	5391811	San Diego	-117.1573	32.7153
24	Tampa	FL	US	1772847910	4174757	Tampa	-82.4584	27.9475
25	Nashville	TN	US	1772848046	4644585	Nashville	-86.7844	36.1659
26	Austin	TX	US	1772847903	4671654	Austin	-97.7431	30.2672
27	Honolulu	HI	US	1772848189	5856195	Honolulu	-157.8583	21.3069

	City	State	Country	Dt	Id	Name	Coord.Lon	Coord.Lat
28	Portland	OR	US	1772848200	5746545	Portland	-122.6762	45.5234
29	St. Louis	MO	US	1772848189	4407066	St Louis	-90.1979	38.6273
30	Raleigh	NC	US	1772847991	4487042	Raleigh	-78.6386	35.7721
31	Sacramento	CA	US	1772848111	5389519	Sacramento	-121.3177	38.4666
32	New Orleans	LA	US	1772848041	4335045	New Orleans	-90.0751	29.9547
33	San Juan	PR	US	1772848204	3837213	San Juan	-68.5251	-31.5371
34	Kansas City	MO	US	1772847938	4393217	Kansas City	-94.5786	39.0997
35	San Jose	CA	US	1772848089	5392171	San Jose	-121.8950	37.3394
36	San Antonio	TX	US	1772847996	4726206	San Antonio	-98.4936	29.4241
37	Fort Myers	FL	US	1772848164	4155995	Fort Myers	-81.8406	26.6217
38	Santa Ana	CA	US	1772848207	5392900	Santa Ana	-117.8678	33.7456
39	Oakland	CA	US	1772848208	5378538	Oakland	-122.2708	37.8044
40	Indianapolis	IN	US	1772848116	4259418	Indianapolis	-86.1580	39.7684
41	Cleveland	OH	US	1772848139	5150529	Cleveland	-81.6954	41.4995

	City	State	Country	Dt	Id	Name	Coord.Lon	Coord.Lat
42	Pittsburgh	PA	US	1772847657	5206379	Pittsburgh	-79.9959	40.4406
43	Cincinnati	OH	US	1772848098	4508722	Cincinnati	-84.4569	39.1620
44	Columbus	OH	US	1772847974	4509177	Columbus	-82.9988	39.9612
45	West Palm Beach	FL	US	1772847997	4177887	West Palm Beach	-80.0534	26.7153
46	Jacksonville	FL	US	1772848212	4160021	Jacksonville	-81.6556	30.3322
47	Ontario	CA	US	1772848213	5379439	Ontario	-117.6509	34.0633
48	Kahului	HI	US	1772848214	5847411	Kahului	-156.4700	20.8947
49	Burbank	CA	US	1772848214	5331835	Burbank	-118.3090	34.1808
50	Hartford	CT	US	1772847981	4835806	Hartford	-72.7662	41.7668
51	Charleston	SC	US	1772847951	4574324	Charleston	-79.9309	32.7766
52	Milwaukee	WI	US	1772848169	5263058	Milwaukee	-87.9668	43.0006
53	Anchorage	AK	US	1772848217	5879400	Anchorage	-149.9003	61.2181
54	Albuquerque	NM	US	1772848218	5454711	Albuquerque	-106.6511	35.0845
55	Omaha	NE	US	1772848218	5074472	Omaha	-95.9378	41.2586

	City	State	Country	Dt	Id	Name	Coord.Lon	Coord.Lat
56	Buffalo	NY	US	1772847984	5110629	Buffalo	-78.8784	42.8865
57	Boise	ID	US	1772848001	5586437	Boise	-116.2035	43.6135
58	South San Francisco	CA	US	1772848220	5397765	South San Francisco	-122.4078	37.6547

Display First 10 Records of Population Dataset

The following code displays the **first 10 rows** of the `population_df` dataset:

- This allows you to quickly inspect the data and understand its structure.
- You can see sample values for columns such as city names, population counts, and any other relevant fields.
- Using `.head(10)` is helpful for verifying that the dataset has loaded correctly and contains the expected information.

```
In [73]: # Print the Population dataset to display the first 10 records
population_df.head(10)
```

Out[73]:

	INDEX	TABLE_ID	CITY	SUMMARY_LEVEL	PLACE_FIPS	GEO_ID	STATE	ST
0	0	100001	Coffman Cove	160	16360	16000US0216360	AK	
1	1	100002	Cohoe	160	16420	16000US0216420	AK	
2	4	100005	College	160	16750	16000US0216750	AK	
3	5	100006	Cooper Landing	160	17190	16000US0217190	AK	
4	6	100007	Copper Center	160	17300	16000US0217300	AK	
5	7	100008	Cordova	160	17410	16000US0217410	AK	
6	8	100009	Covenant Life	160	17670	16000US0217670	AK	
7	9	100010	Craig	160	17740	16000US0217740	AK	
8	10	100011	Crooked Creek	160	17850	16000US0217850	AK	
9	11	100012	Crown Point	160	17960	16000US0217960	AK	



Display Final Dataset

The following code displays the **final dataset** containing all the cleaned and merged records.

- Using `final_df` allows us to quickly inspect the data after all transformations and merges are complete.
- This helps verify that all columns and rows are correctly combined and ready for analysis.
- Viewing the dataset ensures data integrity before performing further calculations or visualizations.

```
In [74]: # Print the Final dataset to display all the records
final_df
```

Out[74]:

	Rank	Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
0	1	Hartsfield Jackson Atlanta International Airport	ATL	Atlanta	Metro Atlanta	GA	52511402
1	2	Dallas Fort Worth International Airport	DFW	Dallas	Dallas-Fort Worth	TX	42351316
2	3	Denver International Airport	DEN	Denver	Greater Denver	CO	40012895
3	4	O'Hare International Airport	ORD	Chicago	Chicago	IL	38575693
4	5	Los Angeles International Airport	LAX	Los Angeles	Greater Los Angeles	CA	37760834
5	6	John F. Kennedy International Airport	JFK	New York City	New York Metro	NY	31466102
6	7	Charlotte Douglas International Airport	CLT	Charlotte	Greater Charlotte	NC	28523822
7	8	Harry Reid International Airport	LAS	Las Vegas	Las Vegas Valley	NV	28244966
8	9	Orlando International Airport	MCO	Orlando	Greater Orlando	FL	27859783
9	10	Miami International Airport	MIA	Miami	Miami Metro	FL	26588002
10	11	Phoenix Sky Harbor International Airport	PHX	Phoenix	Metro Phoenix	AZ	25595723
11	12	Seattle Tacoma International Airport	SEA	Seattle	Seattle Metro	WA	25414592
12	13	San Francisco International Airport	SFO	San Francisco	San Francisco Bay Area	CA	25078968

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
13	14	Newark Liberty International Airport	EWR	Newark	New York Metro	NJ	24544320
14	15	George Bush Intercontinental Airport	IAH	Houston	Greater Houston	TX	23349157
15	16	Logan International Airport	BOS	Boston	Greater Boston	MA	21090721
16	17	Minneapolis Saint Paul International Airport	MSP	Minneapolis	Minneapolis–Saint Paul	MN	18054481
17	18	Fort Lauderdale Hollywood International Airport	FLL	Fort Lauderdale	Miami Metro	FL	17096131
18	19	Laguardia Airport	LGA	New York City	New York Metro	NY	16715567
19	20	Detroit Metropolitan Airport	DTW	Detroit	Detroit Metro	MI	16110696
20	21	Philadelphia International Airport	PHL	Philadelphia	Philadelphia Metro	PA	15102261
21	22	Salt Lake City International Airport	SLC	Salt Lake City	Wasatch Front	UT	13543570
22	23	Baltimore Washington International Airport	BWI	Baltimore	Baltimore metropolitan area	MD	13221461
23	24	Dulles International Airport	IAD	Washington, D.C.	Washington Metro	VA	13003234
24	25	San Diego International Airport	SAN	San Diego	Greater San Diego	CA	12780013
25	26	Ronald Reagan Washington National Airport	DCA	Washington, D.C.	Washington Metro	VA	12750892

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
26	27	Tampa International Airport	TPA	Tampa	Tampa Bay area	FL	12075591
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33	34	St. Louis Lambert International Airport	STL	St. Louis	St. Louis	MO	7807362
34	35	Raleigh Durham International Airport	RDU	Raleigh	Research Triangle	NC	7584394
35	36	William P. Hobby Airport	HOU	Houston	Houston	TX	7116967
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47	48	Pittsburgh International Airport	PIT	Pittsburgh	Pittsburgh	PA	4862376
48	49	Cincinnati Northern Kentucky International Air...	CVG	Cincinnati	Cincinnati	OH	4500414
49	50	John Glenn Columbus International Airport	CMH	Columbus	Columbus	OH	4387395

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
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52	53	Ontario International Airport	ONT	Ontario	Greater Los Angeles	CA	3494554
53	54	Kahului Airport	OGG	Kahului	Maui	HI	3415615
54	55	Hollywood Burbank Airport	BUR	Burbank	Greater Los Angeles	CA	3383237
55	55	Hollywood Burbank Airport	BUR	Burbank	Greater Los Angeles	CA	3383237
56	56	Bradley International Airport	BDL	Hartford	Hartford	CT	3285194
57	57	Charleston International Airport	CHS	Charleston	Charleston	SC	3115194
58	58	Milwaukee Mitchell International Airport	MKE	Milwaukee	Milwaukee	WI	3104054
59	59	Ted Stevens Anchorage International Airport	ANC	Anchorage	Anchorage	AK	2767856
60	60	Albuquerque International Sunport	ABQ	Albuquerque	Albuquerque	NM	2690218
61	61	Eppley Airfield	OMA	Omaha	Omaha	NE	2583212
62	62	Buffalo Niagara International Airport	BUF	Buffalo	Buffalo–Niagara Falls	NY	2499295

	Rank	Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
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64	1	John F. Kennedy International Airport	JFK	New York City	Queens, New York	NY	34765387
65	2	Los Angeles International Airport	LAX	Los Angeles	Los Angeles, California	CA	23438418
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68	5	Newark Liberty International Airport	EWR	Newark	Newark, New Jersey	NJ	14690370
69	6	O'Hare International Airport	ORD	Chicago	Chicago, Illinois	IL	13865434
70	7	Hartsfield Jackson Atlanta International Airport	ATL	Atlanta	College Park, Georgia	GA	13813088
71	8	Dallas Fort Worth International Airport	DFW	Dallas	Grapevine, Texas	TX	11841330
72	9	George Bush Intercontinental Airport	IAH	Houston	Houston, Texas	TX	11675134
73	10	Washington Dulles International Airport	IAD	Washington, D.C.	Dulles, Virginia	VA	10060851
74	11	General Edward Lawrence Logan International Airport	BOS	Boston	Boston, Massachusetts	MA	8491518

Rank		Airport	IATA_Code	City	Metro_Area	State	Passengers_2024
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76	13	Fort Lauderdale Hollywood International Airport	FLL	Fort Lauderdale	Fort Lauderdale, Florida	FL	6897993
77	14	Seattle Tacoma International Airport	SEA	Seattle	Seatac, Washington	WA	6056779
78	15	Charlotte Douglas International Airport	CLT	Charlotte	Charlotte, North Carolina	NC	4623143
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80	17	Philadelphia International Airport	PHL	Philadelphia	Philadelphia, Pennsylvania	PA	3704176
81	18	Harry Reid International Airport	LAS	Las Vegas	Paradise, Nevada	NV	3592102
82	19	Daniel K. Inouye International Airport	HNL	Honolulu	Honolulu, Hawaii	HI	3533266
83	20	Detroit Metropolitan Wayne County Airport	DTW	Detroit	Romulus, Michigan	MI	3353501
84	21	Minneapolis Saint Paul International Airport	MSP	Minneapolis	Saint Paul, Minnesota	MN	3320850
85	22	Phoenix Sky Harbor International Airport	PHX	Phoenix	Phoenix, Arizona	AZ	2821736
86	23	Laguardia Airport	LGA	New York City	Queens, New York	NY	1686395

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87	24	Luis Muñoz Marín International Airport	SJU	San Juan	San Juan, Puerto Rico	PR	1458172
88	25	Tampa International Airport	TPA	Tampa	Tampa, Florida	FL	1384192
89	26	Antonio B. Won Pat International Airport	GUM	Tamuning	Tamuning, Guam	Guam	1342793
90	27	Salt Lake City International Airport	SLC	Salt Lake City	Salt Lake City, Utah	UT	1311617
91	28	Baltimore Washington International Airport	BWI	Baltimore	Linthicum Heights, Maryland	MD	1156882
92	29	Austin Bergstrom International Airport	AUS	Austin	Austin, Texas	TX	1064668
93	30	San Diego International Airport	SAN	San Diego	San Diego, California	CA	995537